NutriScan: AI-Powered Food Detection

Milestone 1 — Data Understanding, Exploratory Analysis and Food Localisation

Prepared for the Chief Operating Officer and Data Science Manager · 31 October 2026 · Capstone Project

1. The Business Problem

Maintaining a balanced diet depends on knowing what you actually eat, and today that means logging meals by hand. It is slow, it is easy to get wrong, and most people abandon it within weeks. The data that does get recorded is unreliable: self-reported calorie intake is systematically under-estimated, which undermines every recommendation built on top of it.

NutriScan removes the typing. A user photographs a meal and the system identifies the food and returns its nutritional profile immediately. Delivering that requires two capabilities in sequence: **detection** — finding where the food is in a photograph that also contains a table, cutlery and hands — and **classification** — naming the dish inside that region. **This milestone delivers the first of those, and the clean dataset the second one will be trained on. No classifier has been trained yet.**

Pain point today	What NutriScan changes	Business value
Manual logging is slow; users quit	One photograph, instant result	Higher engagement and retention
Self-reported intake is inaccurate	Image-based identification	Trustworthy data for recommendations
Generic dietary advice	Personalisation from logged habits	Premium-tier differentiation
Health data sits in silos	Integration with health apps	Partnerships and ecosystem reach

90.1%	1,700	17	92.0%
FOOD LOCATED	TRAINING IMAGES BUILT	FOOD CATEGORIES	BOXES CORRECT ON REVIEW

2. The Data

Source	Food-101 subset
Total images	16,256
Food categories	17
Corrupt or unreadable files	0
Exact duplicate files removed	0
Images after cleaning	16,256
Images processed this milestone	1,700
Typical image size	512 × 512 px

The library is 16,256 photographs of plated dishes, organised one folder per food type, with the folder name acting as the label. Every file opened correctly and no exact duplicates were found — a cleaner starting point than is typical. Duplicate checking matters commercially as well as technically: had the same photograph appeared in both the training and the test set, our accuracy figures would have been inflated and we would have over-promised on what the product can do.

Detection was run on a stratified random sample of **1,700 images** (100 per category). Section 4 explains why, and what it costs us.

3. What the Data Tells Us

3.1 The categories are not evenly represented

Imbalance ratio (full dataset)	3.91×
Smallest category	apple_pie — 256 images
Largest categories	1,000 images each
Imbalance within our working sample	1.00×

One category, *apple pie*, holds roughly a quarter of the images the others do. Left unaddressed, a model trained on this data learns that guessing apple pie is rarely worth the risk — it will score well on average while being quietly useless for that one food. Any user logging apple pie gets a wrong calorie count.

> **Decision:** Milestone 2 will train with balanced class weights, so errors on scarce categories cost the model more, and will be judged on **macro-F1** — which averages performance across categories equally — rather than plain accuracy, which a majority-biased model can inflate.

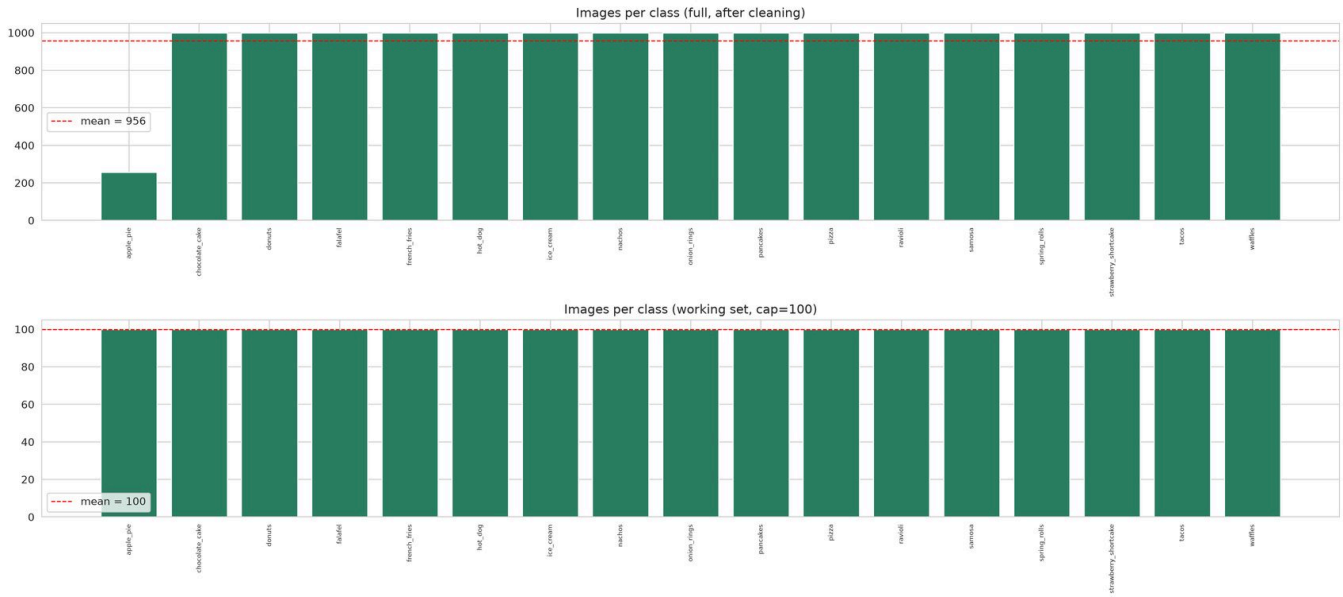


Figure 1. Images per category, full dataset (top) and the working sample (bottom). **What this tells us:** the shortfall is confined to a single category; capping our sample at 100 per category makes the working set look balanced, which is why the full-data ratio is the one quoted above.

3.2 The photographs vary widely — and that is useful



Figure 2. One representative photograph per category. **What this tells us:** several categories are visually close — slice-shaped desserts, stacked discs with syrup, golden fried parcels — and these are where the classifier will struggle.

Most photographs show a single dish, but plates, cutlery, table surfaces and occasionally packaging appear in frame, and lighting, camera angle and plating vary considerably. That variation is an asset: a model trained only on studio-perfect images would fail on the arm's-length photographs real users take.

3.3 Image quality is good, which is itself a risk

Dark images	2.8%
Over-exposed images	0.1%
Low-contrast images	1.2%
Blurry images	5.0%

Only a small fraction of images are poorly lit or out of focus, and the dataset's average colour profile is close to that of ImageNet — the reference collection our pretrained models were built on — which confirms those models can be used without recalibration.

> **The risk this creates:** our training images are *easier* than the ones users will actually submit. Milestone 2 must deliberately degrade training images — varying brightness, contrast and focus — so the model is robust to real conditions. This is a requirement, not a refinement.

4. Finding the Food in the Photograph

4.1 The obstacle, and how we got around it

The dataset tells us *what* is in each photograph but never *where*. Conventional object detectors must be taught from thousands of hand-drawn boxes, which we do not have and which would be expensive and slow to produce.

We avoided that cost entirely by using a **zero-shot detector** — a model that has already learned to associate images with English descriptions. We simply ask it, in plain English, to find "a plate of food", and it returns the region. No annotation, no training, and it generalises to food types it has never been shown.

> **Commercial significance:** this removes a data-labelling project from the critical path. The same approach extends to new cuisines and menu items without commissioning fresh annotation each time.

4.2 Calibrating the confidence threshold

The detector scores its confidence in each region. Set the bar too high and genuine meals are rejected; too low and we accept boxes drawn around tables and plate rims. We tested a range of settings and selected the strictest one that still located food in at least 90% of photographs.

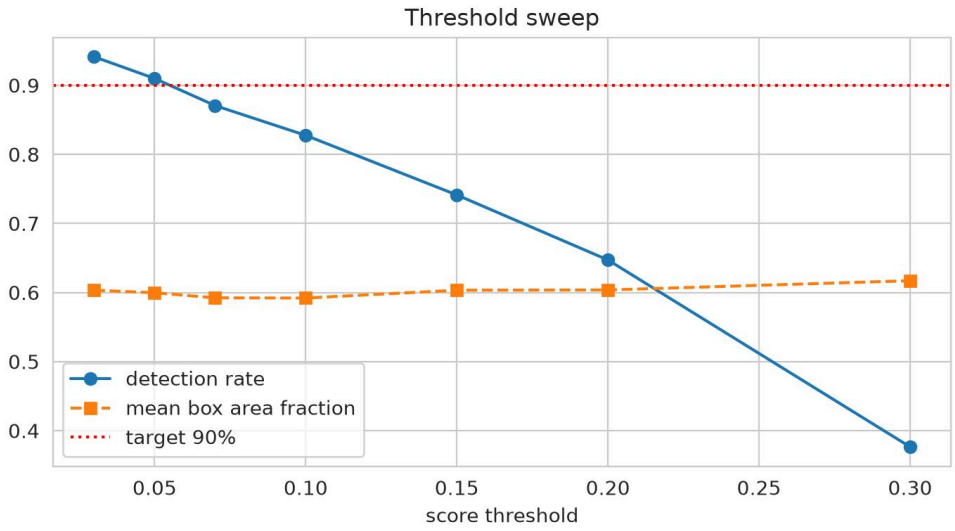


Figure 3. Detection success rate against confidence threshold. **What this tells us:** the chosen setting of 0.05 is the strictest value that still clears the 90% business target.

4.3 Results

Photographs where food was located	**90.1%** (95% confidence interval ±1.4 points)
Business target	90% — **met**
Fell back to using the full photograph	9.9%
— no region scored above the threshold	129 images (7.6%)
— best region was too small to be credible	40 images (2.4%)
Average detector confidence	0.294
Median region size, as a share of the photo	66%
Photographs containing two or more food regions	58.5%

We met the target. Just as importantly, the system never discards an image: when no confident region is found it keeps the whole photograph, so every picture still yields a usable training sample and the pipeline cannot silently lose data.

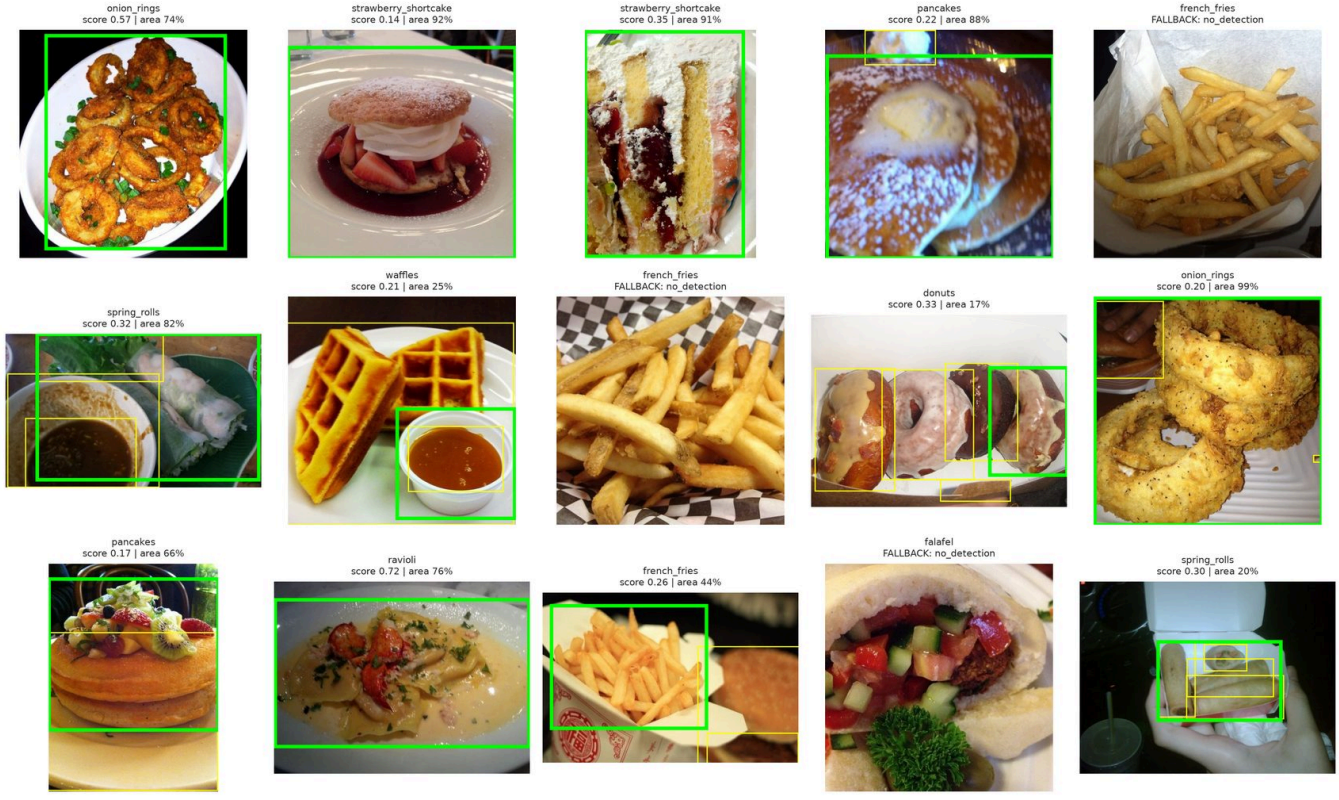


Figure 4. Detected regions on sample photographs; the selected region is outlined in green. **What this tells us:** the region generally sits on the food itself, but on plates holding several items it often encloses only one of them.

4.4 How we verified accuracy honestly

Because no human ever marked the “correct” region for these photographs, there is no automatic score to compute. We therefore reviewed a random sample of detections by eye and judged each one correct, partially correct, or wrong. The result: **92.0% of boxes judged correct and 96.0% judged usable (correct or partially correct) across 50 hand-reviewed detections**.

This is a sample-based estimate, not an exact figure, and we state it as such. Section 6 describes how to make detection quality properly measurable.

4.5 From photograph to model-ready data

Each detected region is cut out with a small margin, squared off so the food is not stretched out of shape, resized to a uniform 224×224 pixels, and colour-standardised to match what pretrained models expect. The results are divided into training, validation and test sets with category proportions held constant, and we verified that no photograph appears in more than one set.



Figure 5. Original photograph with the selected region (top) and the final standardised crop (bottom). **What this tells us:** the pipeline produces consistent, food-centred inputs from inconsistent source photographs.

Training images	1,189
Validation images	255
Test images	256
Total prepared	**1,700**
Smallest per-category test set	15 images
Overlap between sets	None — verified

5. Insights and Recommendations

5.1 The single biggest weakness: meals with more than one item

58.5% of photographs contain two or more distinct food regions, but the system currently keeps only the best one. On a plate holding several items it crops one of them and discards the rest.

> For a calorie-counting product this is the most consequential finding in this report. A user photographing a mixed plate would have only part of their meal counted, and the error is always in the same direction — under-counting. The fix is inexpensive, because every candidate region is already stored: we change which ones we keep, not how we find them.

5.2 Where detection struggles

Situation	Why it happens	What we will do
Loose, scattered foods — *french fries* had the highest fallback rate at 58.0%	No single shape encloses a pile of separate strips	Union of regions rather than one box
Several items on one plate	Only the single best region is kept	Keep every confident region and classify each
Close-ups where food fills the frame	No region stands out from its surroundings	Acceptable — the whole photograph *is* the food
Region includes plate rim or sauce	Prompts are deliberately generic	Tune the wording; raise the minimum size filter
No ground-truth regions exist	Dataset was never annotated with boxes	Hand-label a sample to obtain a true, comparable accuracy score

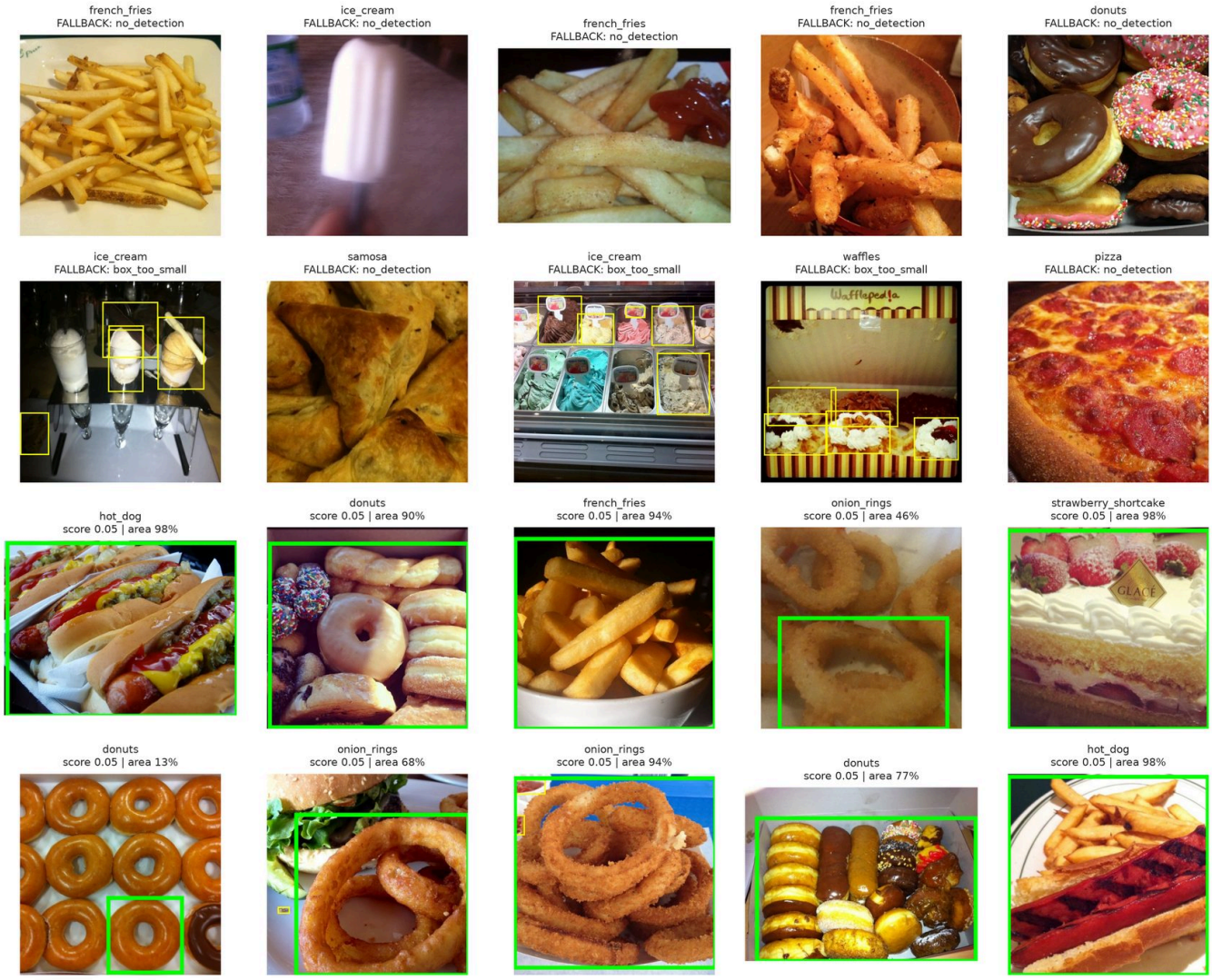


Figure 6. Fallback cases and the lowest-confidence accepted regions. **What this tells us:** most failures are close-ups where keeping the full photograph is the right answer, rather than genuine mistakes.

5.3 Did cropping actually add value?

The median detected region covers 66% of the photograph, which tells us most images in this dataset are *already* close-ups of a single plated dish, framed by a photographer. Cropping therefore changes these particular images less than one might expect.

We report that plainly because it does not make the detection stage redundant — it makes the dataset unrepresentative. The product's real input is a cluttered table photographed at arm's length. Detection is what makes that input tractable, and this dataset simply under-represents the hard case. **Recommendation: validate Milestone 2 on genuine user photographs before launch.**

5.4 Categories we expect to be confused

From visual inspection, the groups most at risk are **chocolate cake / strawberry shortcake / apple pie**, **pancakes / waffles**, **samosa / spring rolls**, and **french fries / onion rings**. These should be the first place we look in the Milestone 2 confusion matrix. They carry a direct business cost: confusing a light dessert with a rich one is a logging error of several hundred calories, which is precisely the kind of inaccuracy that erodes user trust.

6. Plan for Milestone 2

Step	Approach
Starting point	The 1,700 standardised images built in this milestone
Baseline	A small network trained from scratch, for a reference score
Main approach	Transfer learning — EfficientNet-B0, ResNet50, MobileNetV2 for mobile deployment
Why transfer learning	Re-uses a model already trained on millions of photographs, so far less data and time is needed
Robustness	Augmentation — flips, crops, brightness, contrast and blur, to mimic real user photographs
Imbalance	Balanced class weights and label smoothing
How we judge it	Macro-F1 and per-category scores alongside top-1 accuracy; confusion matrix reviewed for the look-alike groups
Final selection	Best validation macro-F1, with a speed and model-size check for mobile; the test set is used once
Beyond Milestone 2	Map categories to nutrition data, estimate portion size, generate recommendations, integrate with health apps

Risks

- **Visually similar foods will cap accuracy.** Some categories are genuinely hard to separate from a photograph alone.
- **Coverage is limited to 17 categories.** Real users eat far more variety; expanding the menu is a data-collection problem, not a modelling one.
- **A photograph cannot reveal hidden ingredients or portion size.** Cooking oil, added sugar and serving weight are invisible, so calorie figures will always carry error. This should be communicated honestly in the product.
- **Dataset photographs are easier than user photographs.** Validate on real submissions before launch.

Infrastructure recommendation

> This milestone ran on a CPU-only machine, where locating food takes about one second per photograph — several hours for the full library, against roughly ten minutes on standard cloud GPU hardware. That constraint is manageable now. It will not be in Milestone 2, where model training rather than detection becomes the dominant cost and CPU-only execution would be impractical. **We recommend provisioning GPU compute before the classification phase begins.**

7. Conclusion

We understood the data, established that it is clean but unevenly distributed across categories, and demonstrated that food can be located automatically in photographs **without commissioning any manual annotation** — meeting the

90% target at 90.1%, with 92.0% of boxes judged correct and 96.0% judged usable (correct or partially correct). From that we built a clean, leak-free dataset of 1,700 standardised food images ready for classification.

We also identified the most important weakness early: 58.5% of photographs contain more than one food item, and the current pipeline keeps only one. Fixing that is inexpensive and is the first item of work in the next phase. The project is on track for Milestone 2.